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CODE

```
#1)Import        NumPy import
numpy as np
print("Assignment 5")

#2)create one dimantional array
print("Create one domentional array") arr =
np.array([10, 20, 30, 40, 50]) print("Array:",
arr) print("\n")

#3)Find Size and Dimension of an Array
print("Size and dimention of array")
print("Number of elements:", arr.size)
print("Dimension:", arr.ndim) print("\n")

#4)Create 2 dimentional array
print("Create 2 dimentional array") arr
= np.array([[10, 20, 30],
            [40, 50, 60]])
print("2-D Array:") print(arr)
print("\n")
```

```
#3)Basic Array Operations print("Basic operations
on array") a = np.array([10, 20, 30, 40, 50])
print("Array:", a) print("Addition:", a
+ 5) #addition print("Subtraction:", a - 5)
#subtraction print("Multiplication:", a * 2)
#multiplication print("Division:", a / 2)
#division print("\n")

#4) Array Indexing print("Indexing of
array") print("First element:", a[0])
print("Third element:", a[2])
print("Last element:", a[-1])
print("\n")

#5)Array Slicing print("Slicing of array")
print("First three elements:", a[0:3])
print("Elements from index
2:", a[2:]) print("\n")

#6)Mathematical and Statistical Operations print("Mathematical and
statistical operations") print("Marks:", a) print("Sum:", np.sum(a))
print("Mean:", np.mean(a)) print("Maximum:", np.max(a))
print("Minimum:", np.min(a)) print("\n")
```

```
#complete program # Create an array
marks = np.array([65, 70, 75, 80, 85])

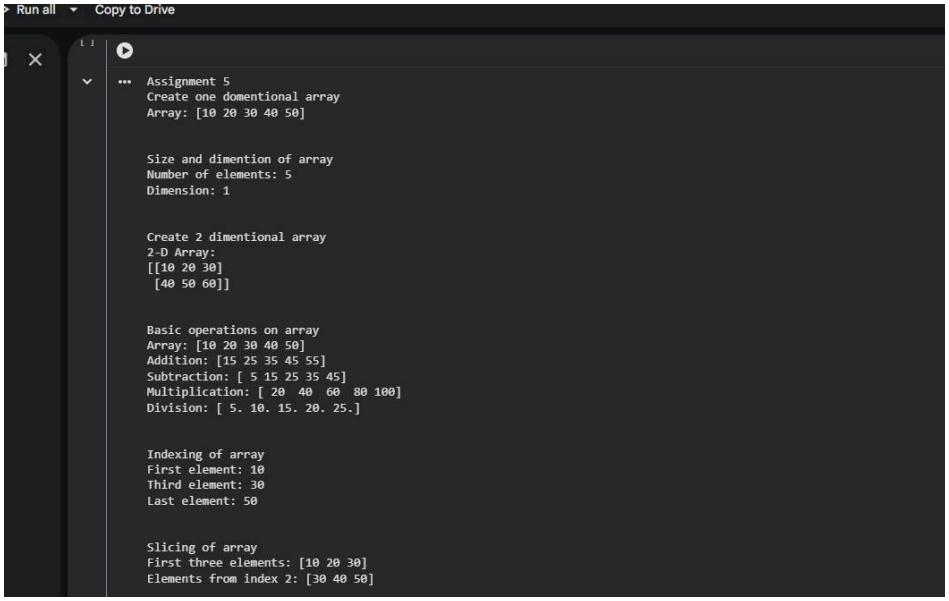
# Display array print("Marks:",
marks)

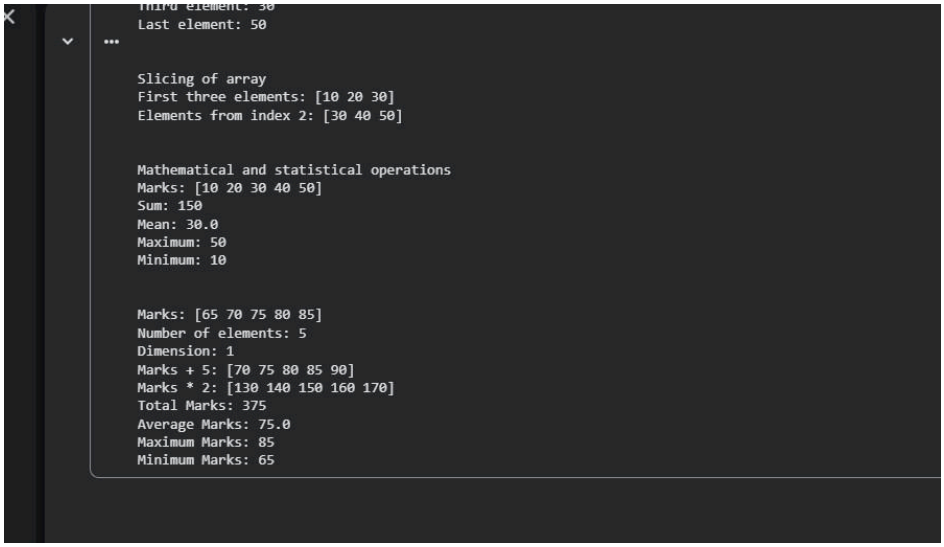
# Display array information print("Number
of elements:", marks.size)
print("Dimension:", marks.ndim)

# Basic operations print("Marks +
5:", marks + 5) print("Marks *
2:", marks * 2)

# Statistical operations print("Total
Marks:", np.sum(marks)) print("Average
Marks:", np.mean(marks)) print("Maximum
Marks:", np.max(marks)) print("Minimum
Marks:", np.min(marks))
```

OUTPUT


Run all
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Assignment 5
Create one domentional array
Array: [10 20 30 40 50]
Size and dimention of array
Number of elements: 5
Dimension: 1
Create 2 dimentional array
2-D Array:
[[10 20 30]
[40 50 60]]
Basic operations on array
Array: [10 20 30 40 50]
Addition: [15 25 35 45 55]
Subtraction: [5 15 25 35 45]
Multiplication: [20 40 60 80 100]
Division: [5. 10. 15. 20. 25.]
Indexing of array
First element: 10
Third element: 30
Last element: 50
Slicing of array
First three elements: [10 20 30]
Elements from index 2: [30 40 50]


Third element: 30
Last element: 50
Slicing of array
First three elements: [10 20 30]
Elements from index 2: [30 40 50]
Mathematical and statistical operations
Marks: [10 20 30 40 50]
Sum: 150
Mean: 30.0
Maximum: 50
Minimum: 10
Marks: [65 70 75 80 85]
Number of elements: 5
Dimension: 1
Marks + 5: [70 75 80 85 90]
Marks * 2: [130 140 150 160 170]
Total Marks: 375
Average Marks: 75.0
Maximum Marks: 85
Minimum Marks: 65